

Creepage and Clearance

CREEPAGE AND CLEARANCE

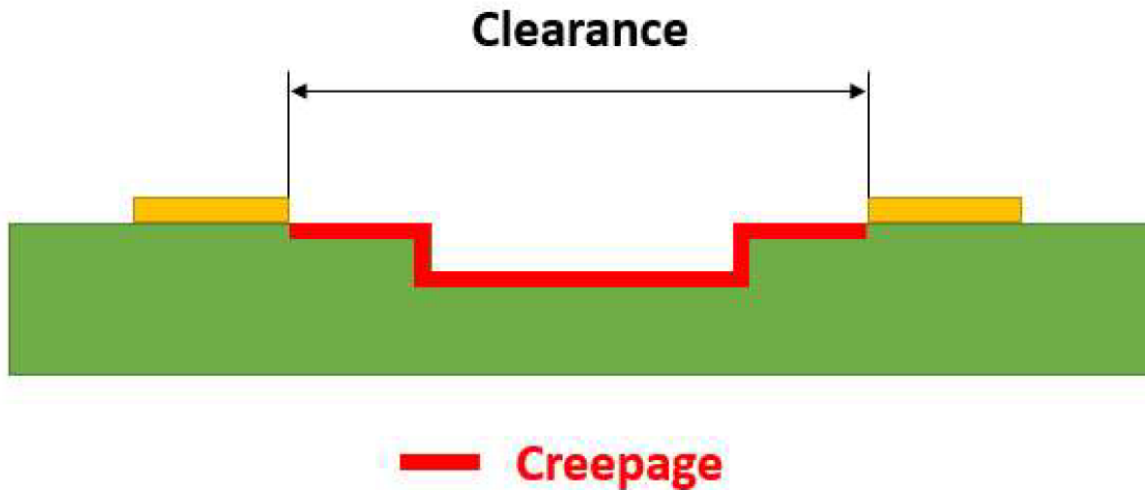
Creepage and clearance are safety-related distances between conductors in a PCB layout, defined differently based on how the distance is measured.

Clearance is the shortest distance through air between two conductive parts. It depends on voltage difference, whether conductors are internal or external, and environmental factors such as humidity, altitude, dust, and conformal coating. These factors influence air breakdown voltage and the risk of arcing.

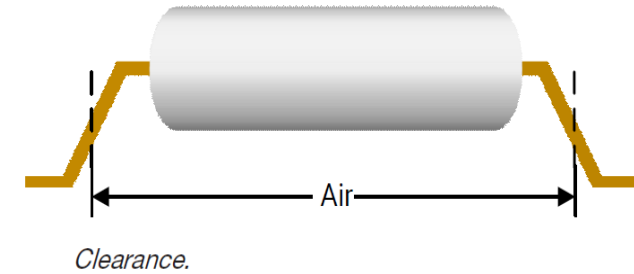
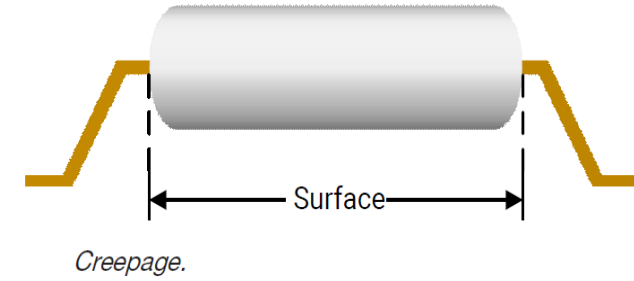
Creepage is the shortest distance along the surface of the insulating material between conductors. It is affected by the board material and environmental conditions, especially moisture and contamination, which can create conductive paths over time.



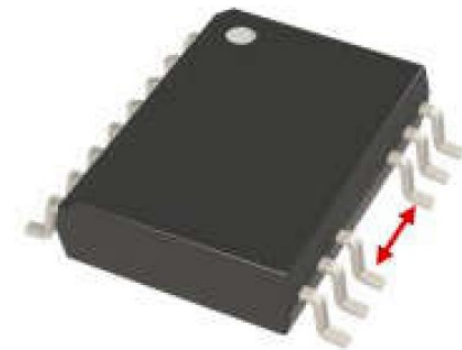
CREEPAGE AND CLEARANCE



Creepage and Clearance on PCB



Creepage and Clearance on Electronic Component



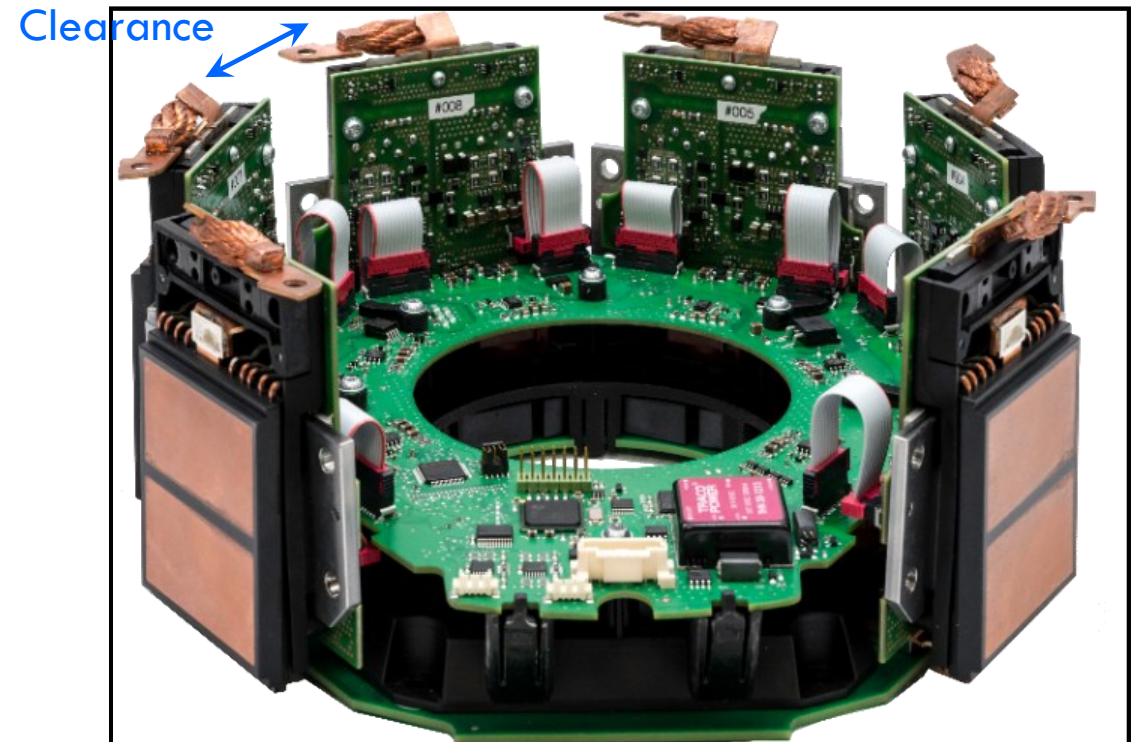
Creepage and Clearance are the same
(component soldered on PCB)

CREEPAGE AND CLEARANCE

High-voltage PCB spacing and IC package selection have become critical factors in achieving compact, high power-density designs while ensuring compliance with safety standards and design requirements. As systems become smaller and more integrated, reduced PCB area and tighter component placement make creepage and clearance constraints increasingly design-limiting, particularly in high-voltage applications.



400 V Battery Management System



400 V Wheel Hub Inverter

MATERIAL AND COMPARATIVE INDEX (CTI)

The Comparative Tracking Index (CTI) classifies insulating materials according to their resistance to surface tracking under electrical stress. The CTI value is determined through a standardized test in which a specified voltage is applied across the material surface while 50 drops of a conductive solution (water containing 0.1% ammonium chloride) are introduced.

The CTI rating corresponds to the highest voltage the material can withstand during this test without sustained tracking, defined as a tracking current exceeding 0.5 A. Based on the measured CTI value, insulating materials are assigned to material groups. These material groups are used in safety standards to determine the required creepage distance for a given insulation level, as part of the methodology for defining creepage, clearance, and high-voltage PCB spacing requirements. Most FR4 materials used in PCB fabrication are rated for material group IIIa.

Material group	CTI range (V_{RMS})
I	$600 \leq CTI$
II	$400 \leq CTI < 600$
IIIa	$175 \leq CTI < 400$
IIIb	$100 \leq CTI < 175$ Or if not specified

POLLUTION DEGREE

Another key parameter in determining required creepage and clearance distances is the pollution degree of the operating environment. Pollution degree classifies the level of environmental contamination that may affect insulation performance and surface conductivity. Four categories are defined:

Pollution Degree 1

No pollution or only dry, nonconductive contamination is present. Equipment is typically sealed to prevent dust and moisture ingress, or the PCB is protected with conformal coating to avoid humidity- or temperature-induced condensation.

Pollution Degree 2

Only nonconductive pollution is normally present, but temporary conductivity may occur due to occasional condensation. Typical examples include laboratories, offices, and enclosed equipment such as servers, telecommunications systems, and wireless infrastructure.

Pollution Degree 3

Conductive pollution is present, or nonconductive pollution may become conductive due to expected condensation. This category includes many industrial environments, agricultural equipment, and unheated factory areas.

Pollution Degree 4

Persistent conductivity exists due to conductive dust, rain, splashing water, or other wet conditions. This classification generally applies to outdoor environments exposed directly to weather.



CREEPAGE ACCORDING TO IEC 60664-1

V_{RMS}	Creepage distances to avoid failure caused by tracking (mm)			
	Pollution degree 1	Pollution degree 2		
	All material groups	Material group		
		I	II	III
63	0.2	0.63	0.9	1.25
400	1.0	2.0	2.8	4.0
800	2.4	4.0	5.6	8.0
1,000	3.2	5.0	7.1	10.0

Creepage is specified using RMS (or DC working) voltage because it concerns surface tracking along insulating materials. Tracking is a time-dependent degradation process caused by continuous electrical stress combined with humidity and contamination. Therefore, creepage distance is determined by the long-term operating voltage rather than short transient peaks.

CLEARANCE ACCORDING TO IEC 60664-1

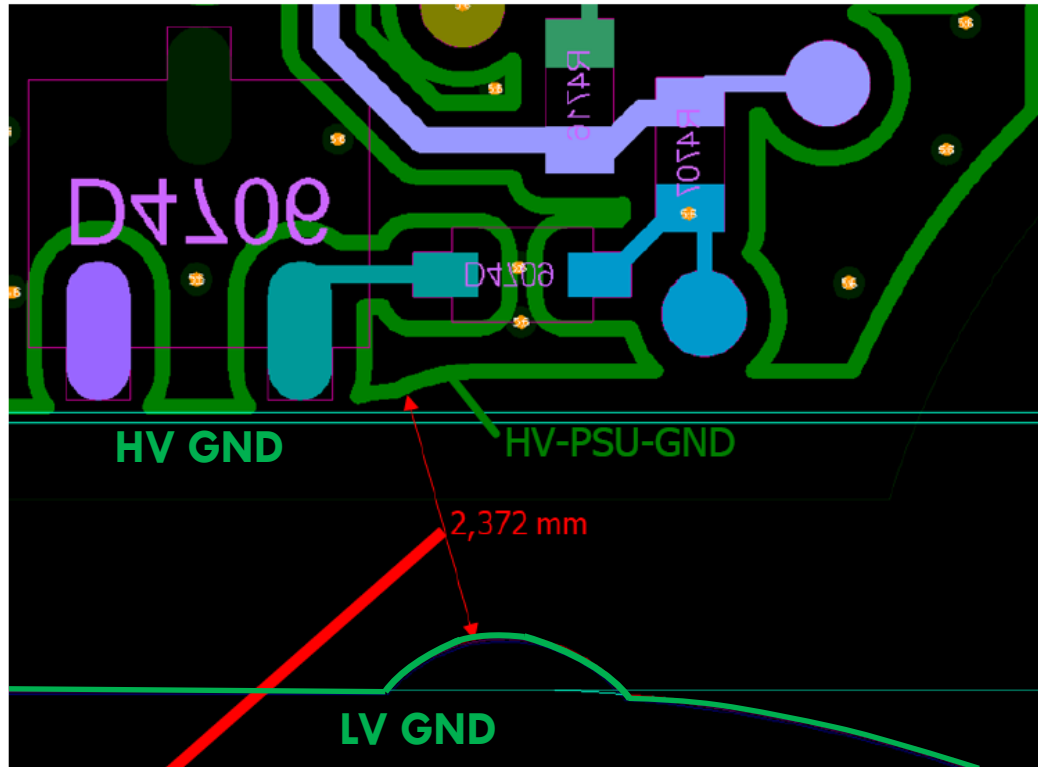
Voltage line to neutral (V_{RMS})	Mains transient/rated impulse voltage (V_{PEAK})			
	Overvoltage category			
	I	II	III	IV
≤ 50	330	500	800	1,500
≤ 150 (for example, 120 V in the U.S.)	800	1,500	2,500	4,000
≤ 300 (for example, 230 V in the European Union, China)	1,500	2,500	4,000	6,000
≤ 600 (for example, industrial motors, ship power)	2,500	4,000	6,000	8,000

Required impulse withstand voltage (kV)	Minimum clearances (mm)		
	Pollution degree		
	1	2	3
0.5	0.04	0.2	0.8
1.5	0.5 (0.76)		0.8
2.5	1.5 (1.8)		
4.0	3.0 (3.8)		
6.0	5.5 (7.9)		

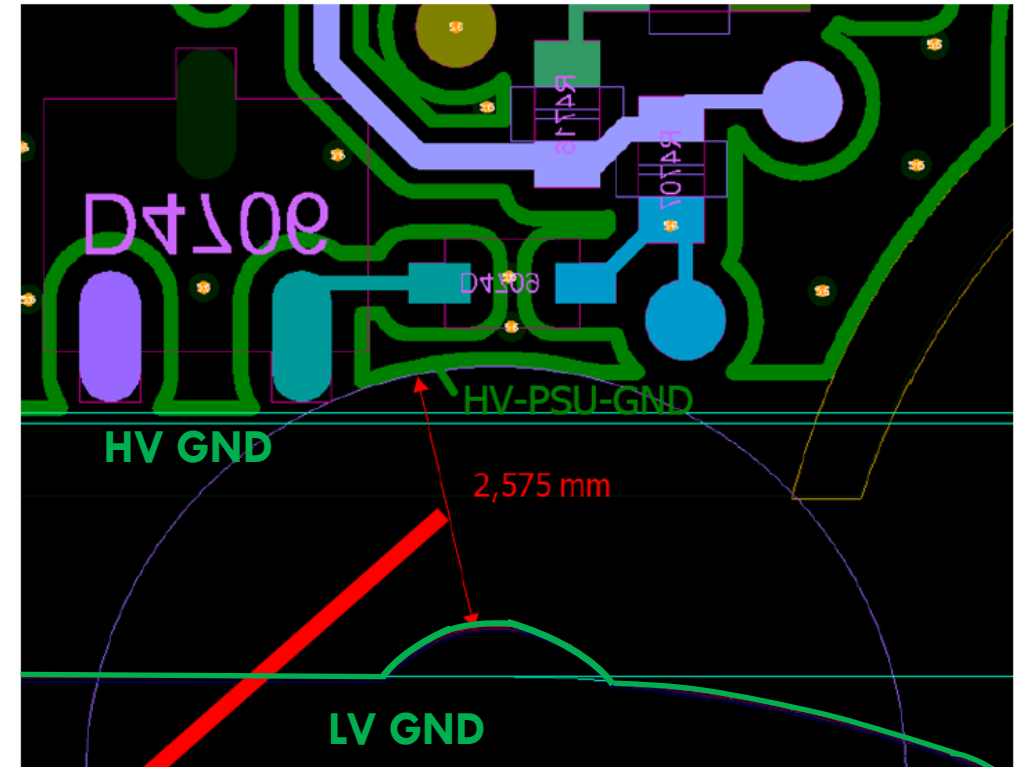
Clearance is specified based on impulse withstand voltage because it relates to air breakdown, which is primarily triggered by short-duration, high-peak transient voltages such as lightning surges or switching spikes.

CREEPAGE ACCORDING TO IEC 60664-1

Some PCB Layout Examples



Before



After

$V_{RMS} = 800 \text{ V}$, Pollution Degree 1

EXAMPLE

Example Scenario: 400 V DC Bus – Industrial DC-DC Converter

- DC bus voltage: 400 VDC
- Overvoltage category: III
- Pollution degree: 2
- PCB material: FR-4
- CTI: 175 → Material Group IIIa

What are the required minimum clearance and creepage distances between primary and secondary?

Step 1: Clearance (based on impulse voltage)

For a 400 V system in Overvoltage Category III, IEC 60664-1 typically assigns:

Required impulse withstand voltage ≈ 6 kV

For Pollution Degree 2, the corresponding clearance is approximately: ≈ 5.5 mm



EXAMPLE

Step 2: Creepage (based on working voltage)

Working voltage = 400 VDC

Pollution degree = 2

Material group = IIIa

From standard tables, the required creepage distance is approximately: ≈ 4.0 mm



References

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- Analysis and Design of Analog Integrated Circuits”, Gray Mayer et al, 5th Edition, Wiley, 2009
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- Power Topologies Handbook, Markus Zehendner, Matthias Ulmann, Texas Instruments, 2020